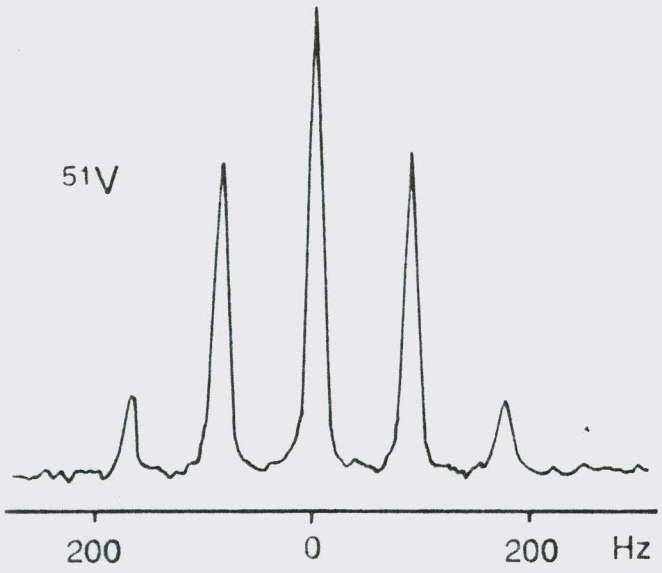
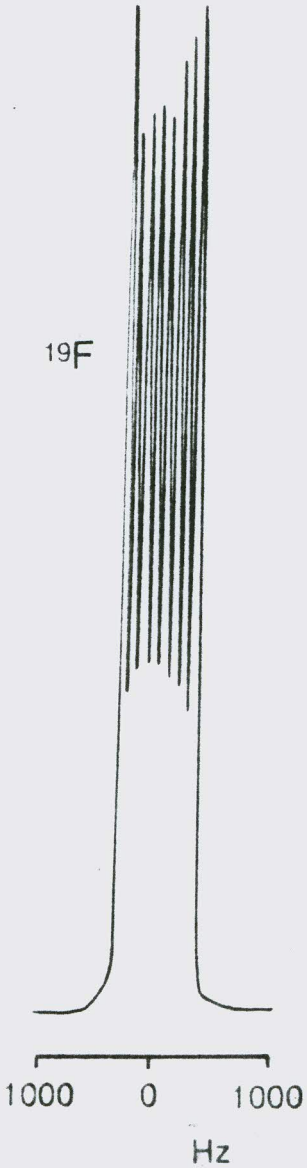
Physical Chemistry Cume
Spectroscopy

1. The V^{51} and F^{19} spectra of the complex ion $[VOF_4]^-$ are shown below. Explain the origin of the number of lines in the V^{51} and F^{19} spectra. What is the structure of the ion as shown by the NMR spectra. V^{51} (I=7/2) and F^{19} (I=1/2)





2. Explain why the deuterium decoupled H^1 spectrum of the deuteriated cyclohexane, $C_6D_{11}H_1$,is a single sharp line at room temperature and two singlets at 170 K.

3. Predict the spectrum (number of lines and relative intensities) of the elements in the ammonium ions and NH_3 shown below:

(A) $[N^{14}H_4]^+$ N^{14} (I=1) H^1 (I=1/2)

(B) $[N^{14}D_4]^+$ D (deuterium (I=1)

(C) $[N^{15}H_4]^+$ N^{15} (I=1/2)

(D) $[N^{15}D_4]+$

(E) $N^{14}H_3$

(F) $N^{15}D_3$

(G) In what ways are the spectra of nitrogen in (A) and (E) different?